Artificial Pancreas (Closed-Loop Glucose Control) — Assignment Brief

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Problem

You will design a **closed-loop insulin delivery controller** for a Type-1 diabetic patient (an *artificial pancreas*). The goal is to keep **blood glucose** $G(t)$ near a safe target despite meal disturbances, using only **ODE-based modeling and control**.

At $t = 10\,\text{min}$ a meal occurs that raises glucose for five minutes. Your controller must meet the performance, actuator, robustness, and safety specifications below.

Model (Input–Output ODEs, Bergman Minimal Model)

We adopt a simplified glucose–insulin interaction model:

$$\dot{G}(t) = -k_1(G(t) - G_b) - X(t)\,G(t) + D(t), \tag{1}$$

$$\dot{X}(t) = -k_2\,X(t) + k_3(I(t) - I_b), \tag{2}$$

$$\dot{I}(t) = -n(I(t) - I_b) + \gamma\,u(t), \tag{3}$$

where:

- $G(t)$ = blood glucose concentration (mg/dL) — **measured output** $y(t) = G(t)$.
- $I(t)$ = plasma insulin concentration (mU/L).
- $X(t)$ = insulin action on glucose uptake (1/min).
- $u(t)$ = **insulin infusion rate** (mU/min) — **control input** .
- $D(t)$ = exogenous glucose appearance due to meal (mg/dL/min) — **disturbance**.

Nominal parameters. $G_b = 90\,\text{mg/dL},\ I_b = 15\,\text{mU/L},\ k_1 = 0.01,\ k_2 = 0.05,\ k_3 = 5 \times 10^{-4},\ n = 0.1,\ \gamma = 0.01.$

Meal disturbance. $D(t) = \begin{cases} 300\,\text{mg/dL/min}, & 10\,\text{min} \le t < 15\,\text{min}, \\ 0, & \text{otherwise.} \end{cases}$

Reference and measured output. Target glucose (setpoint) $r(t) = 100\,\text{mg/dL}$. Measured output $y(t) = G(t)$. You may add sensor noise in simulation as $y_m(t) = G(t) + n(t)$.

Controller Design Task

Use $u(t)$ (insulin rate) to regulate $G(t)$ near $r(t)$. **Implement the controller as ODEs** (PI / PID with integral state, or state-feedback with integral action). Include anti-windup if saturation is present.

Specifications & Requirements (Pass/Fail)

Simulation setup. Simulate $t \in [0, 100]$ minutes (continuous time). Apply the meal $D(t)$ at $t = 10\,\mathrm{min}$.

Performance Summary

- **Steady-state glucose accuracy:** $|e_{ss}^{step}| = 0$.
- **Post-meal transient:** Peak glucose $\leq 180\,\mathrm{mg/dL}$.
- **Disturbance recovery:** Return to within $\pm 10\,\mathrm{mg/dL}$ of setpoint within $30\,\mathrm{min}$ after the meal ends.
- **Actuator constraint:** $0 \leq u(t) \leq 5\,\mathrm{mU/min}$ (no negative insulin). If saturation occurs, use anti-windup and still meet specs.
- **Robustness:** For $\pm 20\%$ changes in k_1, k_2, n, the closed loop remains stable and meets the above glucose bounds.
- **Safety:** No hypoglycemia: $G(t) \geq 70\,\mathrm{mg/dL}$ for all t.

Deliverables

1. Simulink model of the ODE plant (1)–(3) and the controller (integrators and sums only).
2. Plots of $G(t)$, $I(t)$, and $u(t)$ highlighting peak glucose, recovery time, and any actuator saturation.
3. A results table reporting: steady-state error , peak , time to return to $\pm 10\,\mathrm{mg/dL}$, and $\max u(t)$. Indicate PASS/FAIL.
4. A short discussion of the trade-off between fast regulation and hypoglycemia risk, and how integral action and saturation handling affect performance.

Suggested Results Table (Template)

Scenario	Steady-state Error (mg/dL)	Peak G (mg/dL)	Recovery time (min)	$\max u$ (mU/min)	Pass/Fail
Nominal params					
k_1, k_2, n $+20\%$					
k_1, k_2, n -20%					
Meal only (no control)					

Notes for students: If insulin saturates, anti-windup is expected. Consider reasonable initial conditions $G(0) = G_b$, $I(0) = I_b$, $X(0) = 0$.